JOHNS HOPKINS
WHITING SCHOOL
of ENGINEERING

Computer Science EN.601.439
Machine Learning for Single-cell and Spatial Genomics
Fall 2026

Instructor

Professor Uthsav Chitra, uthsav@jhu.edu, https://uthsavc.github.io/
Office hours: Th 5:45PM-6:45PM
Office hours location: Malone 319

Teaching Assistant

Sayuni (Sai) Dharmasena sdharma5@jhu.edu
Office hours: TBD
Office hours location: Malone 216

Meetings

Gilman 55
Location/Time: T/Th 4:30-5:45PM

Textbook

Suggested texts:

- Alberts et al. Molecular Biology of the Cell, 4th edition. Garland Science (2002).
- Goodfellow, Bengio, and Courville. Deep Learning. MIT Press (2016).
- Peyre and Cuturi. Computational Optimal Transport. ArXiv:1803.00567, 2018.
- Heumos, L., Schaar, A.C., Lance, C. et al. Best practices for single-cell analysis across modalities. Nat Rev Genet (2023). https://doi.org/10.1038/s41576-023-00586-w
- Lahnemann et al. Eleven grand challenges in single-cell data science. Genome Biology (2020).

Online Resources

We will share course materials on Canvas and the course Github: https://github.com/chitra-lab/ML-for-single-cell-and-spatial-genomics-fall-2026

Course Information

- Recent experimental advances enable the measurement of DNA, RNA and other diverse molecular modalities inside individual cells at an unprecedented scale and resolution. Computational and machine learning (ML) methods are essential for analyzing and interpreting these high-dimensional, single-cell genomics datasets.

This course introduces computational/ML frameworks that are often used to analyze modern single-cell and spatial datasets. Topics include but are not limited to: matrix factorization; autoencoders and contrastive learning; graphs and manifold learning; graph neural networks; computational optimal transport (OT).

Prerequisites

Expected course background in probability (i.e. at least one of EN.553.311 Intermediate Probability and Statistics; EN.553.420 Probability; or EN.553.421 Honors Probability), linear algebra, and calculus. A machine learning/data science course is strongly recommended. No biology background is necessary.

Course Goals

Specific Outcomes for this course are that

- Students will learn about fundamental computational frameworks in matrix factorization/dimensionality reduction, graph theory, and optimal transport.
- Students will gain hands-on experience in implementing/applying these machine learning methods to real biological datasets.
- Students will develop scientific communication and presentation skills.

This course will address the following CSAB ABET Criterion 3 Student Outcomes

Graduates of the program will have an ability to:

<u>SO1. Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.</u>

SO2. Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.

<u>SO3. Communicate effectively in a variety of professional contexts.</u>

SO4. Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.

<u>SO5. Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.</u>

SO6. Apply computer science theory and software development fundamentals to produce computing-based solutions.

Tentative Course Schedule

Unit 0: Background on single-cell and spatial genomics
Unit 1: Matrix factorization / linear dimensionality reduction
Unit 2: Non-linear dimensionality reduction (e.g. autoencoders, contrastive learning, foundation models)
Unit 3: Manifold learning / random walks

Unit 4: Clustering/GNNs + spatial methods
Unit 5: Optimal transport (OT)

See class Github repository for schedule.

Course Expectations & Grading
Attendance and class participation are required and are a component of the course grade. Students will complete written and programming homework assignments; regular (bi-weekly) in-class assessments; three midterm exams; and a final project and/or exam.

In-class assessments: 25% (5% each)
Oral exam: 25%
Homework: 10%
Labs: 10%
Attendance and participation: 5%
Final project + oral exam: 25%

Assignments & Readings
In-class assessments: There will be short in-class assessments (quizzes) after every unit. The lowest quiz grade will be dropped.

Oral: There will be a 10 minute oral exam (with either Uthsav or Sai) scheduled around the middle of the semester.

Homeworks: There will be programming and mathematical assignments for every unit, posted on Canvas. Programming assignments will be in Python. AI use is discouraged – it is beneficial for the students to be able to answer homework questions on their own, as questions may appear in in-class assessments or the oral exam.

Labs: There will be in-class labs for every unit focusing on using current single cell/ spatial algorithms on real data. In these labs, the use of AI (LLMs, agents) is explicitly encouraged.

Final: Students have the option of either taking a final exam or completing a final project. For the final project, students may work individually or in groups of 2-3. More guidance on the final project (e.g. appropriate topics) will be given in towards the middle of the semester.

Late Policy
Students each get 5 free late days for homework assignments for use throughout the semester. At most 3 late days can be used on any single assignment, but otherwise can be

used at the student's discretion without asking permission. Beyond these 5 late days, assignments will not be accepted after they are due, so please plan ahead and consider the possibility of unexpected events.

Ethics

The strength of the university depends on academic and personal integrity. In this course, you must be honest and truthful, abiding by the *Computer Science Academic Integrity Policy*:

> Cheating is wrong. Cheating hurts our community by undermining academic integrity, creating mistrust, and fostering unfair competition. The university will punish cheaters with failure on an assignment, failure in a course, permanent transcript notation, suspension, and/or expulsion. Offenses may be reported to medical, law or other professional or graduate schools when a cheater applies.
>
> Violations can include cheating on exams, plagiarism, reuse of assignments without permission, improper use of the Internet and electronic devices, unauthorized collaboration, alteration of graded assignments, forgery and falsification, lying, facilitating academic dishonesty, and unfair competition. Ignorance of these rules is not an excuse.
>
> Academic honesty is required in all work you submit to be graded. Except where the instructor specifies group work, you must solve all homework and programming assignments without the help of others. For example, you must not look at anyone else's solutions (including program code) to your homework problems. However, you may discuss assignment specifications (not solutions) with others to be sure you understand what is required by the assignment.
>
> *If* your instructor permits using fragments of source code from outside sources, such as your textbook or on-line resources, you must properly cite the source. Not citing it constitutes plagiarism. Similarly, your group projects must list everyone who participated.
>
> Falsifying program output or results is prohibited.
>
> Your instructor is free to override parts of this policy for particular assignments. To protect yourself: (1) Ask the instructor if you are not sure what is permissible. (2) Seek help from the instructor, TA or CAs, as you are always encouraged to do, rather than from other students. (3) Cite any questionable sources of help you may have received.
>
> On every exam, you will sign the following pledge: "I agree to complete this exam without unauthorized assistance from any person, materials or device. [Signed and dated]". Your course instructors will let you know where to find copies of old exams, if they are available.

Report any violations you witness to the instructor.

You can find more information about university misconduct policies on the web at these sites:

- For undergraduates: https://studentaffairs.jhu.edu/policies-guidelines/undergrad-ethics/
- For graduate students: http://e-catalog.jhu.edu/grad-students/graduate-specific-policies/

AI policy. Students are not allowed to use AI tools (e.g., ChatGPT, Gemini, Claude) to directly look up the answers to homework problems. Students are allowed to use AI tools to review course materials and for the labs/final project. We emphasize that AI tools may write code or mathematics that is incorrect. Thus, students are responsible for critically assessing the quality and veracity of their work. Being able to critically review the output of AI tools is an increasingly vital skill in modern settings.